

Notification System

Admin manages all notifications from **one screen on the website**.

They do **not** need to open WhatsApp, Postmark, or push provider websites to create templates. Everything is done from the admin panel.

1. Trigger

A **trigger** is **anything** that should cause a notification to go out.

It is not only login or logout. It can be any event or condition on the website, for example:

Trigger example	When it fires
Login	User signs in on the website
Logout	User signs out
Not logged in for 1 day	User has not visited the website for 24 hours
Not logged in for 1 week	User has not visited for 7 days
Password reset	User asks to reset password
Order placed	User completes a purchase

One trigger = one row in the admin table.

You can add as many triggers as the website needs.

2. Channels (where the message goes)

Each trigger can send on **three channels**:

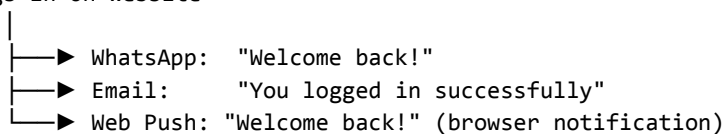
Channel	What the user gets	Service used (in background)
WhatsApp	Message on WhatsApp	WhatsApp Cloud API
Email	Email in inbox	Postmark
Web Push	Pop-up in the browser	Web Push

****Note:**** We use ****Web Push only**** (browser).

When a trigger fires, the website can send on one, two, or all three channels — each with its own template.

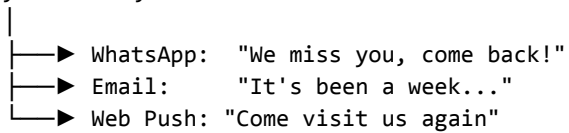
Example — trigger: Login

User logs in on website



Example — trigger: Not logged in for 1 week

User away for 7 days



3. Admin screen (one table)

Trigger	WhatsApp	Email	Web Push
Login	template	template	template
Logout	template	template	template
Not logged in 1 day	template	template	template
Not logged in 1 week	template	template	template

- ****Row**** = one trigger
- ****Column**** = one channel
- ****Each cell**** = one template for that trigger on that channel

Admin can from each cell:

- Create template
- Edit template
- Turn on / off
- Test send

4. Sandbox setup (for candidates — no real API keys)

Candidates will **not** get real WhatsApp Cloud API or production Postmark keys.

Use **sandbox / free accounts** only.

WhatsApp — Sandbox

1. Create a free account on [Meta for Developers](https://developers.facebook.com).
2. Create an app → add ****WhatsApp**** product.
3. Use the ****test phone number**** and ****temporary token**** from API Setup.
4. Add your own phone number to the ****test recipient list**** (only those numbers can receive test messages).
5. Put in `.env``:

...

WHATSAPP_ACCESS_TOKEN=your_test_token

PHONE_NUMBER_ID=your_test_phone_number_id

...

Token expires often — generate a new one from Meta when tests fail.

Email — Postmark (free)

6. Sign up at [Postmark](https://postmarkapp.com).
7. Create a **developer server** — free tier gives **200 emails free** (enough for practice).
8. Verify your email as sender.
9. Put in `.env`:

...

POSTMARKAPP_TOKEN=your_server_token

POSTMARK_FROM_EMAIL=your_verified_email@example.com

...

Web Push — Free tier

10. Sign up at [OneSignal](https://onesignal.com) (free).
11. Create a website app → enable **Web Push** only (skip Android / iOS).
12. Subscribe in your browser to get a **subscription ID** for test sends.
13. Put in `.env`:

...

ONESIGNAL_APP_ID=your_app_id

ONESIGNAL_REST_API_KEY=your_api_key

...

Do not commit `.env` or share tokens.

5. Practice tasks

Task A — Pick any trigger (e.g. Login)

14. Open **Notification Settings** in admin.
15. Find the **Login** row (or any trigger row).
16. Set up all three channels:

Channel	Steps
WhatsApp	Create Template → save → Sync until approved → Test to your phone
Email	Create Template → subject + body → save → Test to your email

Web Push	Create Template → title + body → turn off iOS and Android → save → Test in browser
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✓ Done when you get test message on phone, email, and browser.

Task B — Second trigger (e.g. Logout or Not logged in 1 week)

17. Pick another row (Logout, or inactive-user trigger).
18. Do the same for WhatsApp, Email, Web Push.
19. Use ****different message text**** than Task A.

✓ Done when second row has three working templates.

Task C — Edit and toggle

20. ****Edit**** one template → change text → test again.
21. ****Turn off**** one channel toggle → turn ****on**** again.

✓ Done when you understand edit + on/off.

Task D — Explain (no notes)

Answer in simple words:

22. What is a trigger? Give 3 examples (not only login).
23. What are the three channels?
24. Why create templates in admin panel instead of Postmark / WhatsApp site?
25. What is Web Push?

6. FAQ

Q: Where do I create templates?

A: Only in admin ****Notification Settings****. The system talks to WhatsApp Cloud API, Postmark, and Web Push for you.

Q: Does every trigger send all three channels?

A: Only if all three cells have templates, toggles are ****on****, and the website code fires that trigger.

Q: Can I add a new trigger?

A: Yes. New row → fill WhatsApp, Email, Web Push. Same steps every time.

Q: Real WhatsApp / Postmark account?

A: No. Use ****sandbox**** and ****free tier**** only for practice.

Q: Mobile app push?

A: No. ****Web Push**** in browser only.

7. Assignment — what you must build

Time: 2 days

AI tools: Allowed — use any AI tool you want (ChatGPT, Cursor, Copilot, etc.)

You must build the full notification system on **frontend** and **backend**.

Part	Tech	Host on
Backend	Python + Django	[Render](https://render.com)
Frontend	Your choice (React, Next.js, etc.)	[Vercel](https://vercel.com)

Backend (Django on Render)

- Admin APIs for triggers and templates
- Connect to ****WhatsApp Cloud API**** (sandbox), ****Postmark**** (free), ****Web Push**** (free)
- Fire notifications when a trigger happens on the website
- Store triggers, templates, on/off toggles, variable mappings

Frontend (on Vercel)

- ****Website**** where users can log in, log out, etc. (triggers happen here)
- ****Admin panel**** — one table: rows = triggers, columns = WhatsApp / Email / Web Push
- Create, edit, test send, and turn channels on/off ****from the admin panel only****
- Web Push subscribe in the browser

Minimum triggers to support

At least **2 triggers**, for example:

- Login
- Logout

Or any others from section 1 (e.g. not logged in for 1 day / 1 week).

Each trigger must work on all **3 channels** when toggles are on.

Sandbox only

Use sandbox / free accounts (section 4). No production WhatsApp or Postmark keys.

What to submit

26. ****GitHub repo**** (or two repos — backend + frontend)
27. ****Live backend URL**** (Render)
28. ****Live frontend URL**** (Vercel)
29. Short README: how to log in as admin, which triggers you built, env vars needed

8. Checklist

- ☐ Sandbox WhatsApp set up + test phone added
- ☐ Postmark free account + sender verified
- ☐ Web Push free account + browser subscribed
- ☐ Django backend built and deployed on ****Render****
- ☐ Frontend + admin panel built and deployed on ****Vercel****
- ☐ At least 2 triggers with all 3 channels working
- ☐ Task A — one trigger, all 3 channels tested
- ☐ Task B — second trigger, all 3 channels tested
- ☐ Task C — edited template + used toggle
- ☐ Task D — can explain triggers and channels
- ☐ GitHub + live URLs submitted